

LAB # 1

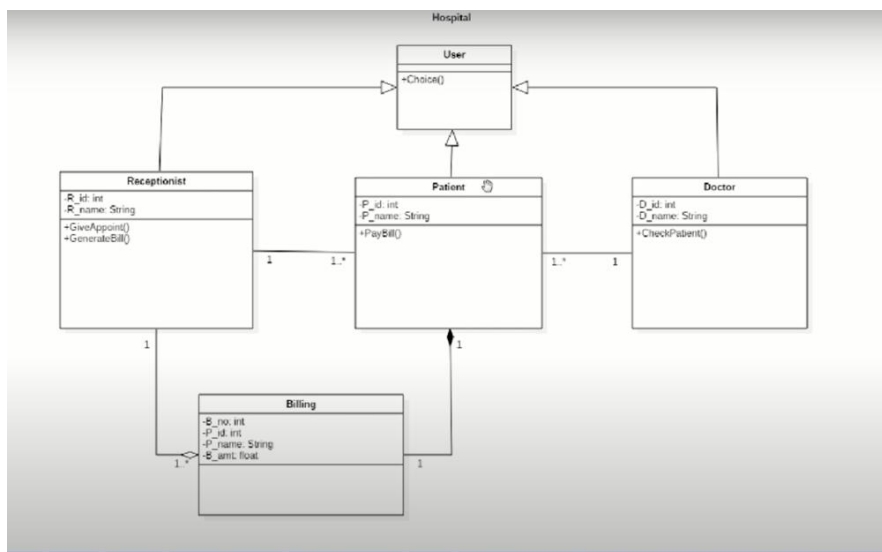
Introduction to Eclipse IDE

OBJECTIVE

Introduction to Eclipse IDE, creating projects, its classes and mapping UML class diagram in to java code.

Lab Task:

Write the java code with the help of the given class diagram of Hospital Management System.



Code:

```

Page x HospitalManagementSystem.java x
rce History
package hospitalmanagementsystem;
class HospitalManagementSystem {
    public static void main(String[] args) {
    }
}

Output - HospitalManagementSystem (run) x Hospital.java x
rce History
class User {
    void choice() {
        System.out.println("Welcome to Hospital System!");
    }
}

class Patient extends User {
    int id;
    String name;
    Patient(int id, String name) {
        this.id = id;
        this.name = name;
    }

    void payBill() {
        System.out.println(name + " has paid the bill.");
    }
}

class Doctor extends User {
    int id;
    String name;

    Doctor(int id, String name) {
        this.id = id;
        this.name = name;
    }

    void checkPatient(Patient p) {
        System.out.println("Dr. " + name + " is checking patient " + p.name);
    }
}

class Receptionist {
    int id;
    String name;
    Receptionist(int id, String name) {
        this.id = id;
        this.name = name;
    }

    void giveAppointment(Patient p, Doctor d) {
        System.out.println("Receptionist " + name + " gives appointment of " + d.name + " to " + p.name);
    }

    Billing generateBill(Patient p, float amount) {
        System.out.println("Bill generated for " + p.name);
        return new Billing(1, p.id, p.name, amount);
    }
}

class Billing {
    int billNo;
    int patientId;
    String patientName;
    float amount;
    Billing(int billNo, int patientId, String patientName, float amount) {
        this.billNo = billNo;
        this.patientId = patientId;
        this.patientName = patientName;
        this.amount = amount;
    }

    void showBill() {
        System.out.println("----- Bill Details -----");
        System.out.println("Bill No: " + billNo);
        System.out.println("Patient: " + patientName);
        System.out.println("Amount: " + amount);
        System.out.println("-----");
    }
}

public class Hospital {
    public static void main(String[] args) {
        User u = new User();
        u.choice();

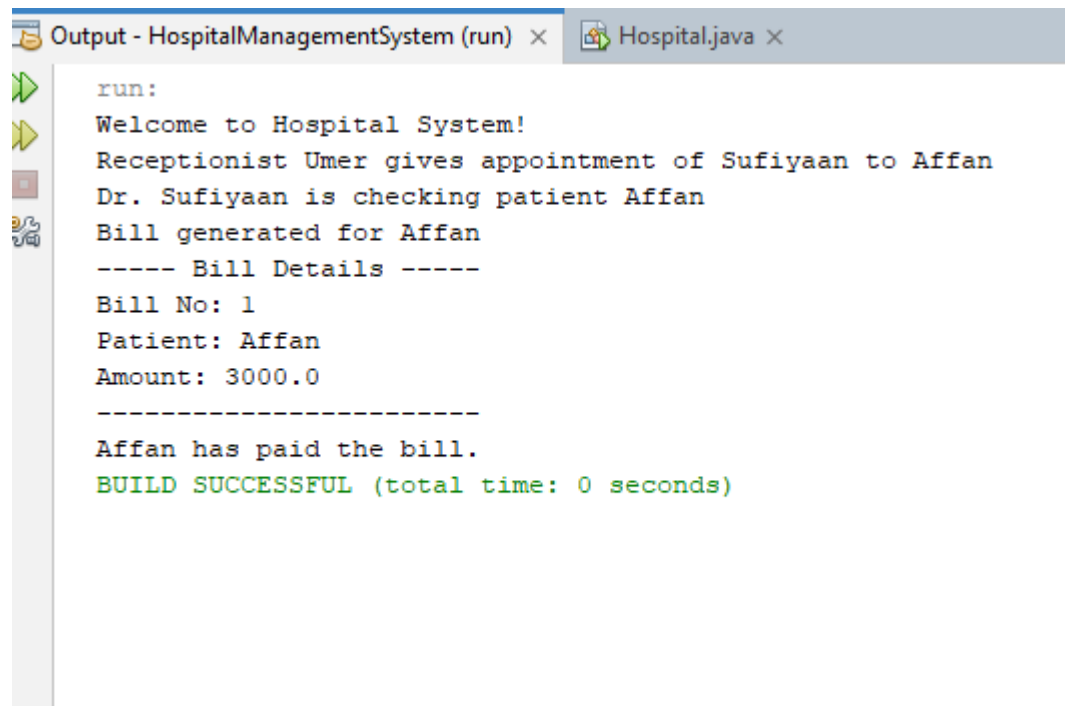
        Patient p1 = new Patient(1, "Affan");
        Doctor d1 = new Doctor(101, "Sufiyyaan");
        Receptionist r1 = new Receptionist(201, "Umer");

        r1.giveAppointment(p1, d1);
        d1.checkPatient(p1);

        Billing b1 = r1.generateBill(p1, 3000);
        b1.showBill();

        p1.payBill();
    }
}

```

Output:

The screenshot shows an IDE window with two tabs: "Output - HospitalManagementSystem (run)" and "Hospital.java". The output window displays the following text:

```
run:
Welcome to Hospital System!
Receptionist Umer gives appointment of Sufiyaan to Affan
Dr. Sufiyaan is checking patient Affan
Bill generated for Affan
----- Bill Details -----
Bill No: 1
Patient: Affan
Amount: 3000.0
-----
Affan has paid the bill.
BUILD SUCCESSFUL (total time: 0 seconds)
```